

ATS Accuracy Metrics

Day 17 – ATS System Evaluation

1. Purpose

This document summarizes the quantitative evaluation metrics generated during Day 17 ATS System Testing.

The ATS-generated candidate decisions were compared against predefined manual review decisions across 12 controlled test cases.

The evaluation measures:

- Accuracy
- Precision
- Recall
- F1 Score
- True Positives
- True Negatives
- False Positives
- False Negatives
- Mismatch count
- Category-level accuracy

2. Evaluation Dataset

Total test cases:

12

Candidate categories:

Tech Role
Non-Tech Role
Fresher Resume
Senior Profile

Job roles tested:

```
Data Scientist
Backend Developer
Frontend Developer
Business Analyst
```

3. Decision Rule

For the Day 17 evaluation, ATS scores were converted into binary decisions using:

```
Score >= 70 → SELECT
Score < 70  → REJECT
```

The positive class used for precision and recall calculation was:

```
SELECT
```

4. Confusion Matrix Results

The evaluation produced:

```
True Positive (TP): 7
True Negative (TN): 4
False Positive (FP): 0
False Negative (FN): 1
```

Confusion matrix:

	Manual SELECT	Manual REJECT
ATS SELECT	7	0
ATS REJECT	1	4

5. Accuracy

Accuracy measures the percentage of all candidate decisions that matched manual review.

Formula:

```
Accuracy = (TP + TN) / Total Cases
```

Calculation:

$$\text{Accuracy} = (7 + 4) / 12$$

$$\text{Accuracy} = 11 / 12$$

$$\text{Accuracy} = 91.67\%$$

Result

91.67%

6. Precision

Precision measures how often candidates selected by the ATS were also selected by manual review.

Formula:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Calculation:

$$\text{Precision} = 7 / (7 + 0)$$

$$\text{Precision} = 100.00\%$$

Result

100.00%

No false-positive selections occurred in the controlled test set.

7. Recall

Recall measures how many manually selectable candidates were successfully identified by the ATS.

Formula:

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Calculation:

$$\text{Recall} = 7 / (7 + 1)$$

$$\text{Recall} = 87.50\%$$

Result

87.50%

The reduction from 100% recall was caused by one false-negative candidate.

8. F1 Score

The F1 Score balances precision and recall.

Formula:

$$\text{F1} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Using:

$$\text{Precision} = 1.00$$

$$\text{Recall} = 0.875$$

produced:

$$\text{F1 Score} = 93.33\%$$

Result

93.33%

9. Overall Metric Summary

Metric	Result
Total Cases	12
Correct Predictions	11
Incorrect Predictions	1
True Positives	7
True Negatives	4
False Positives	0
False Negatives	1
Accuracy	91.67%
Precision	100.00%
Recall	87.50%
F1 Score	93.33%
Mismatch Count	1

10. Category Accuracy

Tech Role

Cases: 3

Correct: 3

Accuracy: 100.00%

Non-Tech Role

Cases: 3

Correct: 2

Accuracy: 66.67%

Fresher Resume

Cases: 3

Correct: 3

Accuracy: 100.00%

Senior Profile

Cases: 3
Correct: 3
Accuracy: 100.00%

Summary:

Category	Cases	Correct	Accuracy
Tech Role	3	3	100.00%
Non-Tech Role	3	2	66.67%
Fresher Resume	3	3	100.00%
Senior Profile	3	3	100.00%

11. Mismatch Metric

Only one mismatch occurred:

TC012

Role: Business Analyst
Profile: Non-Tech Fresher

ATS Score: 69.05
ATS Decision: REJECT

Manual Decision: SELECT

Mismatch rate:

$$1 / 12 \times 100 = 8.33\%$$

Therefore:

Match Rate: 91.67%
Mismatch Rate: 8.33%

12. Metric Interpretation

Precision – 100%

The ATS did not incorrectly select any manually rejected candidate in this controlled dataset.

Recall – 87.50%

The ATS missed one candidate that manual review considered selectable.

Accuracy – 91.67%

Eleven of twelve ATS decisions matched manual decisions.

F1 Score – 93.33%

The combined precision and recall performance remained high within the controlled evaluation set.

13. Limitations

These metrics were calculated from a small controlled dataset containing 12 test cases.

Therefore, the results are useful for:

- Development validation
- Regression testing
- Detecting scoring problems
- Identifying mismatch patterns
- Comparing future system versions

They should not yet be interpreted as statistically representative real-world recruitment performance.

A larger and more diverse human-reviewed dataset would be required for production-level accuracy evaluation.

14. Conclusion

The Day 17 ATS evaluation achieved:

```
Accuracy:  91.67%  
Precision: 100.00%  
Recall:    87.50%  
F1 Score:  93.33%
```

The metrics demonstrate consistent behavior across the majority of the controlled test cases while identifying one false-negative boundary case.

The measurable results provide a baseline that can be used to compare future improvements to the ATS scoring and decision framework.